

Parallel lines



1

- a Their gradients are the same.
- b All graphs have the same angle of inclination or gradient.

2

- a Yes
- b Yes
- c No
- d Yes
- e No
- f No
- g No
- h No

3

- a Yes
- b No
- c Yes
- d Yes
- e No
- f No
- g No
- h No

4

- a No
- b No
- c Yes
- d Yes

5

- a Yes
- b No
- c No
- d Yes

6

- a $-\frac{5}{8}$
- b $-\frac{5}{8}$
- c Yes

7 $y = 2x + 4$

8 $y = 2x + 4$

9

- a $y = 2$
- b $x = -7$
- c $y = -3x - 4$
- d $y = 8x + 5$
- e $y = -2x - 5$

10

- a $b = -\frac{5}{3}$
- b $b = \frac{2}{5}$

11

- a $-\frac{2}{5}$
- b $y = -\frac{2x}{5} + 1$

12

- a -1
- b $y = -x + 18$

13

a -5



b $y = -5x + 40$

Perpendicular lines

14 Two lines that intersect each other at an angle of 90° .

15 -1

16

a No

b No

c Yes

d Yes

e No

f Yes

g Yes

h No

17

a $m = -\frac{36}{5}$

b $m = 10$

18

a $-\frac{1}{6}$

b $-\frac{1}{6}$

c -8

d 1

19

a $-\frac{8}{7}$

b $\frac{8}{7}$

c No

20

a $x = -8$

b $y = -8$

c $y = 2x + 6$

d $y = -\frac{x}{6} + 10$

e $y = -\frac{x}{6} - 4$

21

a $-\frac{1}{5}$

b $y = -\frac{x}{5} + 1$

22

a $\frac{-1}{10}$

b $(5, 10)$

c $x + 10y - 105 = 0$



Applications

23

a -3

b -3

c Yes

24

a $\frac{1}{5}$

b $\frac{1}{5}$

c Yes

d Yes

e Parallelogram

25

a

i 1

ii -1

b Yes, because their gradients multiply to -1 .

26

a

i $m = -1$

ii $m = 1$

iii $m = -1$

iv $m = 1$

b Square

27

a Yes

b No

c Yes